

ЗАДАНИЕ 6

СЛОЖЕНИЕ ЧИСЕЛ С ПЛАВАЮЩЕЙ ЗАПЯТОЙ

Вариант 13

$$A = 25,25, B = 46,55$$

1. Формат Ф1

$$A = (25,25)_{10} = (19,4)_{16} = (0,194)_{16} \cdot 16^2$$

0	1	0	0	0	0	1	0	0	0	0	1	1	0	0	1	0	1	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (46,55)_{10} = (2E,8CCCCD)_{16} = (0,2E8CCCCD)_{16} \cdot 16^2$$

0	1	0	0	0	0	1	0	0	0	1	0	1	1	1	0	1	0	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$1) \quad X_A = 1000010$$

$$X_B = 1000010$$

$$(X_A - X_B)_{\text{пр.}} = 0000000$$

$$(X_A - X_B) = 0; X_C = X_A = X_B = 2$$

а) Оба операнда положительные ($A > 0, B > 0$):

$$2,3) \quad M_A = .000110010100$$

$$M_B = .001011101001$$

$$M_C = .010001111101$$

Результат сложения нормализован.

$$4) \quad M_C = .010001111101$$

0	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 16^{P_c} = (0,47D)_{16} \cdot 16^2 = 71,8125.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 71,8 - 71,8125 = -0,0125$$

$$\delta C = \left| \frac{-0,0125}{71,8} \right| \cdot 100\% = 0,01741\%$$

Результат получился представленным с избытком. Этот факт можно объяснить неточным представлением операндов.

б) $A < 0, B > 0$:

$$2,3) M_B = .001011101001$$

$$M_A = \overline{.000110010100}$$

$$M_C = \overline{.000101010101}$$

Результат вычитания нормализован.

0	1	0	0	0	0	1	0	0	0	0	1	0	1	0	1	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 16^{P_c} = (0,155)_{16} \cdot 16^2 = 21,3125.$$

$$\Delta C = 21,3 - 21,3125 = -0,0125$$

$$\delta C = \left| \frac{-0,0125}{21,3} \right| \cdot 100\% = 0,05869\%$$

с) $A > 0, B < 0$:

$$2,3) M_A = .000110010100$$

$$M_B = \overline{.001011101001}$$

$$M_C = \overline{.111010101011}$$

1	1	0	0	0	0	1	0	0	0	0	1	0	1	0	1	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 16^{P_c} = (-0,155)_{16} \cdot 16^2 = -21,3125.$$

$$\Delta C = -21,3 - (-21,3125) = 0,0125$$

$$\delta C = \left| \frac{0,0125}{-21,3} \right| \cdot 100\% = 0,05869\%$$

2. Формат Ф2

$$A = (25,25)_{10} = (19,4)_{16} = (0,1100101)_2 \cdot 2^5$$

0	1	0	0	0	0	1	0	1	1	0	0	1	0	1	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (46,55)_{10} = (2E,8CCCCD)_{16} = (0,1011101000110011001101)_2 \cdot 2^6$$

0	1	0	0	0	0	1	1	0	0	1	1	1	0	1	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$1) \quad X_A = 10000101$$

$$X_B = 10000110$$

$$(X_A - X_B)_{\text{доп.}} = 11111111$$

$$(X_A - X_B) = -1; X_C = X_B = 6$$

а) Оба операнда положительные ($A > 0, B > 0$):

$$2,3) M_A = .011001010000$$

$$+ \\ M_B = .101110100011$$

$$M_C = 1.000111110011$$

Результат сложения денормализован влево.

$$4) M_C = .100011111001$$

Т.к. выполнен сдвиг мантиссы вправо, характеристику результата нужно увеличить на 1 ($X_C = X_C + 1 = 7$).

0	1	0	0	0	0	1	1	1	0	0	0	1	1	1	1	1	0	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 2^{P_C} = (0,100011111001)_2 \cdot 2^7 = 71,78125.$$

$$\Delta C = 71,8 - 71,78125 = 0,01875$$

$$\delta C = \left| \frac{0,01875}{71,8} \right| \cdot 100\% = 0,02611\%$$

б) A<0, B>0:

$$\begin{array}{r} 2,3) M_B = .101110100011 \\ - \\ M_A = .011001010000 \\ \hline M_C = .010101010011 \end{array}$$

0	1	0	0	0	0	1	0	1	0	1	0	1	0	1	0	0	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 2^{P_c} = (0,10101010011)_2 \cdot 2^5 = 21,29688.$$

$$\Delta C = 21,3 - 21,29688 = 0,00312$$

$$\delta C = \left| \frac{0,00312}{21,3} \right| \cdot 100\% = 0,01467\%$$

с) A>0, B<0:

$$\begin{array}{r} 2,3) M_A = .011001010000 \\ - \\ M_B = .101110100011 \\ \hline M_C = .101010101101 \end{array}$$

1	1	0	0	0	0	1	0	1	0	1	0	1	0	1	0	0	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$C^* = M_C \cdot 2^{P_c} = (-0,10101010011)_2 \cdot 2^5 = -21,29688.$$

$$\Delta C = -21,3 - (-21,29688) = -0,00312$$

$$\delta C = \left| \frac{-0,00312}{-21,3} \right| \cdot 100\% = 0,01467\%$$

В формате Ф2 результаты получились точнее из-за того, что операнды представлены точнее и при нормализации результата сдвиг производился на один двоичный разряд, а не на четыре.

ЗАДАНИЕ 7

УМНОЖЕНИЕ ЧИСЕЛ С ПЛАВАЮЩЕЙ ЗАПЯТОЙ

Вариант 13

$$A = 4,3, B = 0,06$$

1. Формат Ф1

$$A = (4,3)_{10} = (4,4CCCCD)_{16} = (0,44CCCCD)_{16} \cdot 16^1$$

0	1	0	0	0	0	0	1	0	1	0	0	0	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (0,06)_{10} = (0,0F5C29)_{16} = (0,F5C29)_{16} \cdot 16^{-1}$$

0	0	1	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$\text{SignC} = \text{SignA} \oplus \text{SignB}.$$

$$X_A = P_A + d; X_B = P_B + d;$$

$$X_C = X_A + X_B - d;$$

$$P_C + d = \frac{P_A + d + P_B}{P_C} + d - d.$$

$$\begin{array}{r}
 X_A = \quad 1000001 \\
 + \\
 X_B = \quad 0111111 \\
 \hline
 X_A + X_B = 10000000 \\
 - \\
 d = \quad 1000000 \\
 \hline
 X_C = \quad 1000000 \\
 P_C = 0
 \end{array}$$

№	Операнды	СЧП (старшие разряды)	В/СЧП (младшие разряды)	Признак коррекции
0	СЧП	000000000000000000	1111010111000	0
1	0	000000000000000000	-	0
	СЧП	000000000000000000	1111010111000	
	СЧП->2	000000000000000000	001111010111	
2	$[-M_A]_{\text{доп}}$	111101110110011	$-M_A$	1
	СЧП	111101110110011	001111010111	
	СЧП->2	111111011101100	110011110101	
3	$[2M_A]_{\text{пр}}$	000100010011010	$2M_A$	0
	СЧП	000011110000110	110011110101	
	СЧП->2	000000111100001	101100111101	
4	$[M_A]_{\text{пр}}$	000010001001101	M_A	0
	СЧП	000011000101110	101100111101	
	СЧП->2	000000110001011	101011001111	
5	$[-M_A]_{\text{доп}}$	111101110110011	$-M_A$	1
	СЧП	111110100111110	101011001111	
	СЧП->2	111111101001111	101010110011	
6	0	000000000000000000	-	1
	СЧП	111111101001111	101010110011	
	СЧП->2	111111111010011	111010101100	
7	$[M_A]_{\text{пр}}$	000010001001101	M_A	0
	СЧП	000010000100000	111010101100	
	M_C	000010000100000	111010101100	

$$C = (0,420)_{16} \cdot 16^0 = 0,2578125.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 0,258 - 0,2578125 = 0,0001875$$

$$\delta C = \left| \frac{0,0001875}{0,258} \right| \cdot 100\% = 0,07267442\%$$

2. Формат Ф2

$$A = (4,3)_{10} = (4,4CCCCD)_{16} = (0,1000100110011001101)_2 \cdot 2^3$$

0	1	0	0	0	0	0	1	1	0	0	0	1	0	0	1	1	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (0,06)_{10} = (0,0F5C29)_{16} = (0,1111010111)_2 \cdot 2^{-4}$$

0	0	1	1	1	1	1	0	0	1	1	1	0	1	0	1	1	1	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$\begin{array}{r} X_A = 10000011 \\ + \\ X_B = 01111100 \\ \hline X_A + X_B = 11111111 \\ - \\ d = 10000000 \\ \hline X_C = 01111111 \\ P_C = -1 \end{array}$$

№	Операнды	СЧП (старшие разряды)	В/СЧП (младшие разряды)	Признак коррекции
0	СЧП	00000000000000000000	111110101111000	1
	$[-4M_A]_{\text{доп}}$	111101110110011000		
	$[0M_A]_{\text{пр}}$	00000000000000000000		

1	СЧП	111101111011100111000111101010111000	0
	СЧП->4	1111111110111011100111000011000011111010101	
	[2M _A] _{пр}	000010001000110100	
	[4M _A] _{пр}	000100010001101000	
2	СЧП	000111000101111010111000011111010101	1
	СЧП->4	00000001110001011110101110000111111	
	[-M _A] _{доп}	11111101111011001110	
	[0M _A] _{пр}	000000000000000000	
3	СЧП	11111110100111111011010110000111111	0
	СЧП->4	111111111111010011111101010101110000	
	[M _A] _{пр}	00000100010011010	
	СЧП	0000010000100000011110101010110000	

$$C = (0,100001000001)_2 \cdot 2^{-1} = 0,25793457.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 0,258 - 0,25793457 = 0,00006543$$

$$\delta C = \left| \frac{0,00006543}{0,258} \right| \cdot 100\% = 0,02536034\%$$

Погрешности результатов вызваны неточным представлением операндов. В формате Ф2 операнды представлены точнее и погрешность меньше.

ЗАДАНИЕ 8
ДЕЛЕНИЕ ЧИСЕЛ С ПЛАВАЮЩЕЙ ЗАПЯТОЙ
Вариант 13

$$A = 8,2, B = 0,45$$

1. Формат Ф1

$$A = (8,2)_{10} = (8,333333)_{16} = (0,8333333)_{16} \cdot 16^1$$

0	1	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (0,45)_{10} = (0,733333)_{16} = (0,733333)_{16} \cdot 16^0$$

0	1	0	0	0	0	0	0	0	1	1	1	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$X_C = X_A - X_B + d$$

$$d + P_C = \frac{P_A + d - P_B - d}{P_C} + d$$

$$X_C = 1 - 0 + 64 = 65$$

$$P_C = 1$$

№ шага	Действие	Делимое	Частное
0	M_A	0 1 0 0 0 0 0 1 1	0 0 0 0 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 1 0 0 0 1 1 0 1	
	R_0	0 0 0 0 1 0 0 0 0	$R_0 > 0$
	$M_A \rightarrow 4$	0 0 0 0 0 1 0 0 0	0 0 1 1 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 1 0 0 0 1 1 0 1	
	R_0	1 1 0 0 1 0 1 0 1	0 0 1 1 0 0 0 0
1	$\leftarrow R_0$	1 0 0 1 0 1 0 1 0	0 1 1 0 0 0 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_1	1 1 0 0 1 1 1 0 1	0 1 1 0 0 0 0 0
2	$\leftarrow R_1$	1 0 0 1 1 1 0 1 0	1 1 0 0 0 0 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_2	1 1 0 1 0 1 1 0 1	1 1 0 0 0 0 0 0
3	$\leftarrow R_2$	1 0 1 0 1 1 0 1 1	1 0 0 0 0 0 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_3	1 1 1 0 0 1 1 1 0	1 0 0 0 0 0 0 0

4	$\leftarrow R_3$	1 1 0 0 1 1 1 0 1	0 0 0 0 0 0 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_4	0 0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 1
5	$\leftarrow R_4$	0 0 0 1 0 0 0 0 0	0 0 0 0 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 1 0 0 0 1 1 0 1	
	R_5	1 1 0 1 0 1 1 0 1	0 0 0 0 0 0 1 0
6	$\leftarrow R_5$	1 0 1 0 1 1 0 1 0	0 0 0 0 0 1 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_6	1 1 1 0 0 1 1 0 1	0 0 0 0 0 1 0 0
7	$\leftarrow R_6$	1 1 0 0 1 1 0 1 0	0 0 0 0 1 0 0 0
	$[M_B]_{\text{пр}}$	0 0 1 1 1 0 0 1 1	
	R_7	0 0 0 0 0 1 1 0 1	0 0 0 0 1 0 0 1
8	$\leftarrow R_7$	0 0 0 0 1 1 0 1 0	0 0 0 1 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 1 0 0 0 1 1 0 1	
	R_8	1 1 0 1 0 0 1 1 1	0 0 0 1 0 0 1 0

$$C = (0,12)_{16} \cdot 16^2 = 18.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 18,22222222 - 18 = 0,22222222$$

$$\delta C = \left| \frac{0,22222222}{18,22222222} \right| \cdot 100\% = 1,2195122\%$$

2. Формат Ф2

$$A = (8.2)_{10} = (8,333333)_{16} = (0,10000011001100110011)_2 \cdot 2^4$$

0	1	0	0	0	0	1	0	0	0	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$B = (0.45)_{10} = (0,733333)_{16} = (0,111001100110011)_2 \cdot 2^{-1}$$

0	0	1	1	1	1	1	1	1	1	0	0	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$X_C = X_A - X_B + d$$

$$d + P_C = \frac{P_A + d - P_B - d}{P_C} + d$$

$$X_C = 4 - (-1) + 128 = 133$$

$$P_C = 5$$

№ шага	Действие	Делимое	Частное
0	M_A	0 1 0 0 0 0 0 1 1	0 0 0 0 0 0 0 0
	$[-M_B]_{\text{доп}}$	1 0 0 0 1 1 0 1 0	
	R_0	1 1 0 0 1 1 1 0 1	0 0 0 0 0 0 0 0
1	$\leftarrow R_0$	1 0 0 1 1 1 0 1 0	0 0 0 0 0 0 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_1	0 0 0 1 0 0 0 0 0	0 0 0 0 0 0 0 1
2	$\leftarrow R_1$	0 0 1 0 0 0 0 0 0	0 0 0 0 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 0 1 1 0 1 0	
	R_2	1 0 1 0 1 1 0 1 0	0 0 0 0 0 0 1 0
3	$\leftarrow R_2$	0 1 0 1 1 0 1 0 0	0 0 0 0 0 1 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_3	1 1 0 0 1 1 0 1 0	0 0 0 0 0 1 0 0
4	$\leftarrow R_3$	1 0 0 1 1 0 1 0 0	0 0 0 0 1 0 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_4	0 0 0 0 1 1 0 1 0	0 0 0 0 1 0 0 1
5	$\leftarrow R_4$	0 0 0 1 1 0 1 0 0	0 0 0 1 0 0 1 0
	$[-M_B]_{\text{доп}}$	1 0 0 0 1 1 0 1 0	
	R_5	1 0 1 0 0 1 1 1 0	0 0 0 1 0 0 1 0
6	$\leftarrow R_5$	0 1 0 0 1 1 1 0 0	0 0 1 0 0 1 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_6	1 1 0 0 0 0 0 1 0	0 0 1 0 0 1 0 0
7	$\leftarrow R_6$	1 0 0 0 0 0 1 0 0	0 1 0 0 1 0 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_7	1 1 1 1 0 1 0 1 0	0 1 0 0 1 0 0 0
8	$\leftarrow R_7$	1 1 1 0 1 0 1 0 0	1 0 0 1 0 0 0 0
	$[M_B]_{\text{пр}}$	0 1 1 1 0 0 1 1 0	
	R_8	0 1 0 1 1 1 0 1 0	1 0 0 1 0 0 0 1
	$M_C \rightarrow$		0 1 0 0 1 0 0 0 1

$$C = (0,10010001)_2 \cdot 2^5 = 18,125.$$

Определим абсолютную и относительную погрешности результата:

$$\Delta C = 18,22222222 - 18,125 = 0,09722222$$

$$\delta C = \left| \frac{0,09722222}{18,22222222} \right| \cdot 100\% = 0,53353659\%$$

Погрешности результатов вызваны неточным представлением операндов. В формате Ф2 операнды представлены точнее и погрешность меньше.